WORKED SOLUTION

STEP III 2010, Question 11

Mechanics • Friction and Relative Motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

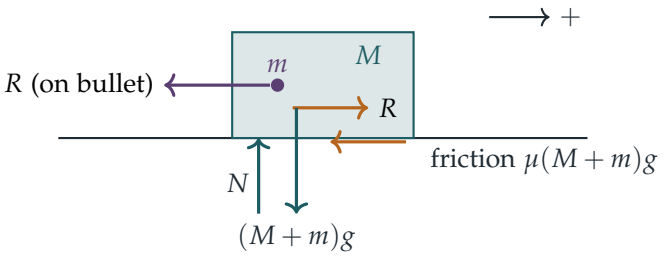
step.maths.org/questions/10-s3-q11

The key idea

Momentum is *not* conserved here, because friction from the ground acts on the block throughout, so the familiar "bullet embeds in block" shortcut is off the table. Instead, treat the bullet and the block as two separate particles, each with its own constant acceleration, linked by the equal and opposite resistance force R. Constant accelerations mean the *suvat* equations do all the work, and the relative motion is just the difference of the two.

Solution

We model the bullet and the block as particles moving along a horizontal line, with the bullet's direction of travel as positive. While the bullet is in the block, the block pushes back on the bullet with force R and, by Newton's third law, the bullet pushes the block forward with force R.



Normal reaction. Vertically nothing accelerates. The bullet is supported by the block, so the ground carries the weight of both: $N = (M+m)g$.

Friction. The forward force R on the block exceeds the greatest friction available, $\mu N = (M+m)\mu g$, since we are told $R > (M+m)\mu g$. So the block does slide, and once it moves friction takes its limiting value $\mu(M+m)g$, acting backwards.

Part (i)

The only horizontal force on the bullet is R backwards, so the bullet decelerates at R/m. The block has net forward force $R-(M+m)\mu g$, so it accelerates forwards at

$$A=\frac{R-(M+m)\mu g}{M}>0.$$

Relative to the block, the bullet's acceleration is $-\frac{R}{m}-A$, so the magnitude of its deceleration relative to the block is

$$a=\frac{R}{m}+\frac{R-(M+m)\mu g}{M},$$

as required.

Part (ii)

Relative to the block, the bullet starts with speed u and decelerates uniformly at a, so it stops relative to the block after time

$$t_1=\frac{u}{a}.$$

At that moment the block's speed is the common speed v. The block started from rest with constant acceleration A, so

$$v=At_1=\frac{Au}{a},\qquad\text{that is,}\qquad av=Au=\frac{Ru-(M+m)\mu gu}{M},$$

which is the required result.

Ask yourself Why not just use conservation of momentum to find v?

Momentum of the bullet–block system is conserved only if no external horizontal force acts. Here the ground's friction acts on the block for the whole time t_1, and removes momentum $(M+m)\mu g\,t_1$. (In a real bullet the time is tiny, so $v\approx mu/(M+m)$, but the question wants the exact value.)

Part (iii)

During time t_1 the block accelerates uniformly from rest to speed v, so its distance is the average speed times the time:

$$s_1=\frac{0+v}{2}\,t_1=\frac{uv}{2a}.$$

Part (iv)

Once the bullet has stopped moving through the block, the two move together as a single body of mass $M+m$. Its weight is still $(M+m)g$, so friction is $\mu(M+m)g$

backwards and the deceleration is μg. Starting at speed v, it slides a further

$$s_2 = \frac{v^2}{2\mu g}$$

before stopping (and then stays at rest, since no horizontal force remains to overcome static friction). The total distance is

$$s_1 + s_2 = \frac{uv}{2a} + \frac{v^2}{2\mu g} = \frac{uv}{2a} + \frac{v}{2\mu g} \cdot \frac{Au}{a} = \frac{uv}{2a\mu g}(\mu g + A),$$

where we used $v = Au/a$ from part (ii) for one factor of v.

Ask yourself How do we get rid of a and A and reach a clean answer?

Put both over the denominator mM and look for a common factor:

$$\mu g + A = \frac{M\mu g + R - (M+m)\mu g}{M} = \frac{R - m\mu g}{M},$$
$$a = \frac{MR + mR - m(M+m)\mu g}{mM} = \frac{(M+m)(R - m\mu g)}{mM}.$$

So $\dfrac{\mu g + A}{a} = \dfrac{m}{M+m}$, the factor $R - m\mu g$ cancelling (it is positive, since $R > (M+m)\mu g > m\mu g$). Hence the total distance moved by the block is

$$\frac{uv}{2\mu g} \cdot \frac{m}{M+m} = \frac{muv}{2(M+m)\mu g},$$

as required.

The case $R < (M+m)\mu g$

Now the forward push R on the block is less than the limiting friction $(M+m)\mu g$, so static friction can hold the block in place: the block never moves. The bullet simply decelerates at R/m inside a stationary block and comes to rest after penetrating a distance $\dfrac{mu^2}{2R}$ (assuming the block is thick enough), and the block stays put throughout.

Checks

- Take $m = 0.02\,\text{kg}$, $M = 2\,\text{kg}$, $u = 300\,\text{m s}^{-1}$, $\mu = 0.5$, $g = 9.8\,\text{m s}^{-2}$ and $R = 1000\,\text{N}$. Then $a \approx 50\,495\,\text{m s}^{-2}$, $v \approx 2.941\,\text{m s}^{-1}$, $s_1 \approx 0.0087\,\text{m}$, $s_2 \approx 0.8827\,\text{m}$, and $s_1 + s_2 \approx 0.8914\,\text{m}$, agreeing with $muv/\big(2(M+m)\mu g\big)$. A step-by-step numerical simulation of the two particles gives the same v and total distance.
- As $R \to \infty$ (an instant collision), $v = Au/a \to mu/(M+m)$, the momentum-conservation answer, as it should.
- Dimensions: $muv/\big((M+m)\mu g\big)$ is $(\text{speed})^2/(\text{acceleration})$, a length.

Common slips

- Using $N = Mg$ for the block, forgetting that the embedded bullet's weight is also carried by the ground.
- Conserving momentum through the impact, which ignores the friction impulse.
- Taking R as the net force on the block, rather than R minus friction.

Beyond STEP

Impulse gives a quick consistency check on the whole motion: the ground's friction impulse over the entire time the block moves must equal the bullet's initial momentum mu, since the system ends at rest. Separating a system into parts, tracking internal forces by Newton's third law and external ones carefully, is the core skill of first-year mechanics, and it leads on to the centre-of-mass frame and inelastic collisions. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas with many problems of exactly this flavour.

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